

Welcome to the Anisotropy Data Challenge

Preparing for IPTA DR3



Anisotropy Data Challenge



Total number of datasets: 12 (6 short + 6 dr3-like)

Short datasets: 50 psrs, Tspan = 10 yrs

dr3-like datasets: 100 psrs, Tspan = 20yrs

pta_1 / pta_1_short
pta_2 / pta_2_short
pta_3 / pta_3_short

- **WN + GWB only datasets:** $\text{Agw} = 2.4\text{e-}15$, observ. freqs = [1400, 2200] MHz (full Tspan coverage)
- the pulsars are the same for all datasets
- **One dataset contains an isotropic GWB, two datasets contain anisotropic GWB.** The two anisotropic realisations are the same obtained from the same SMBHB population and the only difference is the sky location of the dominant binary

pta_4 / pta_4_short

- Same setup as previous dataset. **Anisotropic GWB** which is the same as one of the previous dataset.
- **Red noise** added for all pulsars

pta_hard_1 / pta_hard_1_short
pta_hard_2 / pta_hard_2_short

- One dataset contains an **Anisotropic GWB** (which is the same as the pta_4 dataset) and the other has an **isotropic GWB**.
- **Red Noise and DMvar** added for all pulsars
- Observation frequencies = [800, 1400, 2200] MHz (full Tspan coverage)

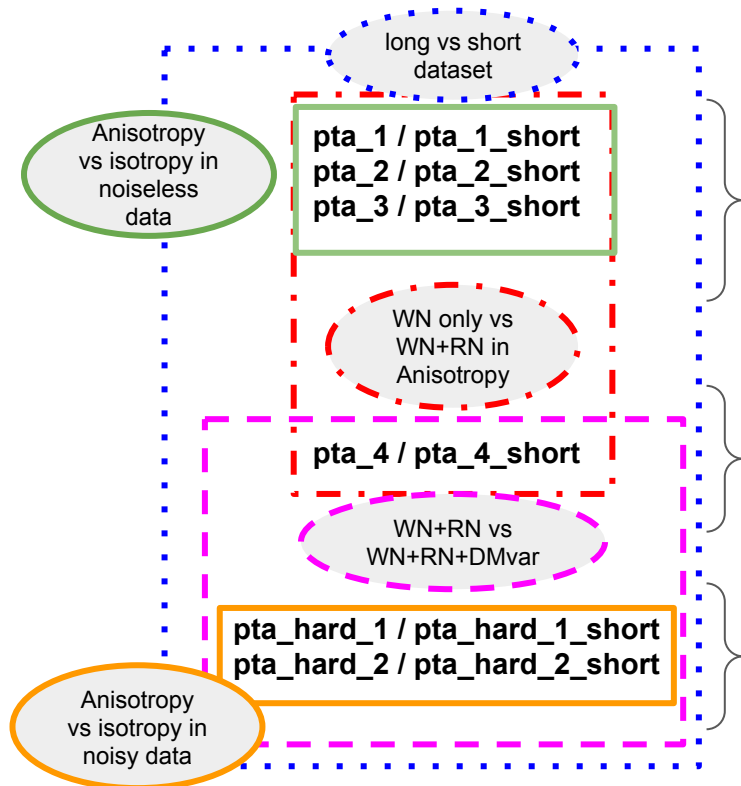
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Dataset Details: Pulsars

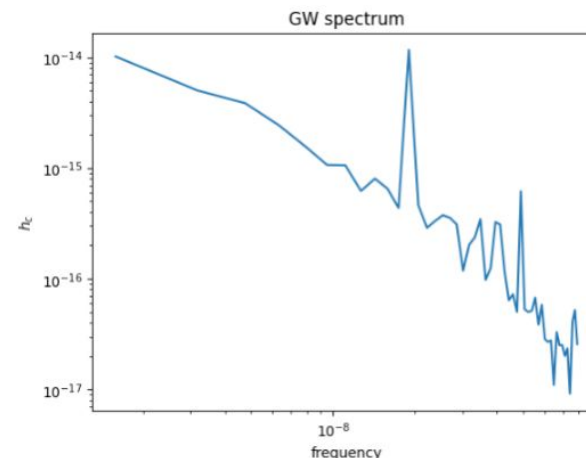


- **Number of pulsars:** 100 (50 for short datasets). The pulsars (intended as sky locations and timing model objects) are the same in all datasets.
- **Tspan:** 20 yrs (10 yrs for short datasets).
- The pulsars are “equally good” timers (same FO and F1, very similar DM) and are uniformly distributed all over the sky.
- **rms:** 1.0 microseconds for all pulsars.
- **Observation frequencies:** [800, 1400, 2200] MHz for the “hard” datasets (which include DMvar), [1400, 2200] MHz for the other datasets. Each frequency band uniformly covers the whole time span.
- **White noise:** $\text{efac} = 1.0$ and $\text{equad} = 1e-6$ for all pulsars.
- **Red Noise** (present in dataset 4 and “hard” datasets) and **DMvar** (present in the “hard” datasets): the hyperparameters of the injected noises are saved in the corresponding *noise_dict.json* file of each dataset. The injected hyperparameters are drawn from the following uniform distributions:
gamma_RN in [2, 4] log10_A_RN in [-17, -14.5]
gamma_DM in [1, 5] log10_A_DM in [-17, -13]

Dataset Details: GWB signal



- The nominal **amplitude** of the injected GWB signal is $2.4\text{e-}15$ (at the frequency of $1/\text{yr-1}$) for all datasets.
- The isotropic GWB realisations were simulated using *createGWB* (from *libstempo.toasim*).
- The anisotropic GWB was computed from a realistic population of SMBHBs, characterised by a very bright and close-by binary that produces a significant peak in the spectrum.



*So... Where is the anisotropy?
How well can your code constrain it?*

Have fun! :)